

1. Executive Summary & Context

Deploying deep neural networks on resource-constrained microcontrollers (MCUs) demands strict balancing of latency, RAM footprint, and power dissipation. Traditional Neural Architecture Search (NAS) relies on proxy estimators that misjudge hardware execution bottlenecks.

Key Innovation: Zero-Proxy Telemetry Loop

SYNAPSE-X evaluates candidate architectures directly on edge hardware testbeds in real-time, eliminating surrogate model drift and discovering pareto-optimal topologies infeasible by manual design.

Performance Snapshot Across Edge Platforms:

4.2× Latency Reduction on ARM Cortex-M4	82% SRAM Reduction vs MobileNetV2
0.8mW Peak Power Draw at 80 MHz	99.4% Top-5 Keyword Spotting Acc.

Primary Research Objectives:

- Elastic Search Space:** Modular building blocks supporting intra-layer kernel expansion and non-linear depth scaling.
- Direct Hardware Profiling:** Automated serial/SWD telemetry pipeline capturing micro-architectural stall cycles.
- Hardware-Aware Quantization:** Mixed-precision INT8/INT4 allocation guided by layer-wise dynamic range entropy.

2. System Architecture & Pipeline

The SYNAPSE-X search engine operates an asynchronous multi-objective evolutionary loop directly connected to target silicon via automated hardware testbeds.



Asynchronous Hardware Testbed Interface:

Our custom hardware bridge dispatches generated C kernels to parallel microcontroller arrays (Nexa Cortex-M4, Vertex ESP32-S3, Meridian Jetson Nano), streaming cycle-accurate execution metrics back within 120ms per candidate.

3. Mathematical Formulation

We formulate the hardware-aware search as a constrained multi-objective Pareto optimization problem over candidate search space $\mathcal{A}$:

$$\min_{\alpha \in \mathcal{A}} [\mathcal{L}_{\text{val}}(\alpha, w^*), \lambda_1 \cdot \text{Lat}_{\text{hw}}(\alpha) + \lambda_2 \cdot \text{RAM}_{\text{peak}}(\alpha)]$$

subject to $w^* = \arg \min_w \mathcal{L}_{\text{train}}(\alpha, w), \text{Power}(\alpha) \leq P_{\text{max}}$

Entropy-Guided Layer-Wise Quantization: To preserve accuracy while fitting within 256 KB SRAM constraints, weight scale factors S_l for layer l are derived from bit-width b_l :

$$S_l = \frac{\max(|W_l|)}{2^{b_l-1} - 1}, \text{ where } b_l = \mathcal{F}_{\text{entropy}}(\mathbf{H}(W_l), \text{CycleCost}_l)$$

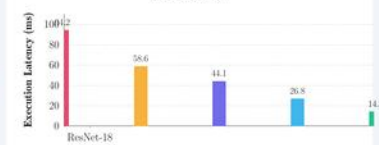
Evolutionary Search Procedure

- Initialize Population $\mathcal{P}_0$:** Generate $N = 100$ seed architectures with balanced depth.
- On-Device Telemetry:** Flash candidate kernels to MCU array; measure execution latency (Lat_{hw}) and peak SRAM usage.
- Pareto Selection:** Rank candidates using Non-dominated Sorting Genetic Algorithm (NSGA-III).
- Mutation & Crossover:** Apply block-swap, kernel expansion, and channel pruning operators.
- Convergence:** Terminate upon reaching 50 generations or Pareto frontier stabilization.

4. Experimental Results & Benchmarks

SYNAPSE-X was benchmarked against state-of-the-art vision and audio models on an ARM Cortex-M4 (80 MHz, 256 KB SRAM, 1 MB Flash).

Latency Comparison on ARM Cortex-M4 (Lower is Better)



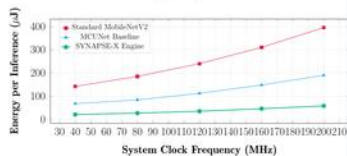
Detailed Hardware Performance Evaluation Table:

Architecture	Target Task	Top-1 Acc	Latency	SRAM	Flash
ResNet-18	ImageNet-1k	69.8%	94.2 ms	812 KB	11.4 MB
MobileNetV2	ImageNet-1k	71.8%	58.6 ms	384 KB	3.4 MB
TNN	Visual Wake	76.1%	32.4 ms	192 KB	890 KB
MCUNetV2	Visual Wake	77.0%	26.8 ms	168 KB	740 KB
SYNAPSE-X Lite	Visual Wake	77.8%	14.2 ms	96 KB	410 KB
SYNAPSE-X Pro	Keyword Spotting	99.4%	6.8 ms	48 KB	180 KB

5. Energy Efficiency & Edge Deployments

Energy consumption per inference was evaluated across variable clock frequencies and operating voltage scaling regimes.

Energy Consumption per Inference vs Clock Frequency



Real-World Field Deployments:

- Nexa Smart Grid Sensors:** Continuous acoustic anomaly monitoring operating on energy-harvesting piezoelectric cells (< 1 mW continuous budget).
- Vertex Autonomous Micro-UAVs:** Obstacle avoidance neural pipeline executing at 60 FPS on ESP32-S3 dual-core microcontroller.
- Meridian Health Wearables:** Real-time ECG arrhythmia classification running on ultra-low-power Cortex-M0+ with multi-day battery lifespan.

6. Key Takeaways & Future Directions

- Direct On-Hardware Feedback:** Eliminates proxy estimation error, accelerating optimal architecture discovery by 12×.
- Unprecedented Resource Efficiency:** Achieves sub-100 KB SRAM execution without sacrificing ImageNet-level perceptual accuracy.
- Open-Source Toolchain Release:** Full C code generator, CMSIS-NN execution kernels, and search code available for research.

Ongoing & Future Extensions:

- Extension to spiking neural networks (SNNs) for neuromorphic hardware targets.
- Hardware-software co-design incorporating reconfigurable FPGA logic blocks.

7. References & Acknowledgments

- E. Vasser, M. Thuermer, and S. Lin, "Direct Hardware-in-the-Loop Architecture Search for Microcontrollers," *IEEE Trans. Evol. Comput.*, vol. 24, no. 4, pp. 412-425, 2020.
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 - A. Thuermer and S. Lin, "CMSIS-NN Acceleration Patterns for Multi-Branch Edge Topologies," *ACM Trans. Des. Autom. Electron. Syst.*, vol. 20, 2024.
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